

All-Optical Wearable Gesture Sensing with a Learnable Photonic Linear Filter Bank for Near-Sensor Classification

Kangjian Di, Jiashu Li, Silin Chen, Shengdi Chen, Fuhao Yu, Nasheeta Mazhar, Wei Jiang, Xuping Zhang, Yixin Zhang, and Ningmu Zou

Abstract—Wearable gesture recognition systems typically rely on multi-channel electrical sensing followed by analog-to-digital conversion, which can incur substantial data movement, latency, and power overhead. Here, we present a photonic feature extraction neural network (PFENN), which provides an electrical-transduction-free optical sensing interface and a photonic preprocessing front end. Gestures are encoded directly in the optical domain via calibrated macro-bending loss in standard single-mode fibers attached to finger joints. The resulting multiple optical intensity streams are processed by a programmable silicon photonic chip that implements a learnable photonic linear filter bank based on a Mach–Zehnder interferometer (MZI) mesh. After this optical linear preprocessing step, a single photodiode and a compact electronic classifier complete recognition. This work demonstrates a training-to-deployment workflow by mapping software-trained weights onto on-chip phase shifters using measured device responses. On a 13-gesture, single-subject dataset (200 samples per class), the system achieves 96.63% accuracy. The system operates without any active electro-optic modulation at the sensing point; on-chip MZMs serve as static, bias-controlled weights during inference. These results support a practical route toward low-overhead near-sensor wearable intelligence by combining passive fiber sensing with learnable photonic linear preprocessing.

Index Terms—Wearable sensing, gesture recognition, optical fiber macro-bending loss, programmable photonics, MZI mesh, near-sensor computing.

I. INTRODUCTION

Wearable hand gesture recognition is a key building block for human–machine interfaces in rehabilitation, assistive technologies, robotics, and extended reality. Recent advances in machine learning and wearable sensing have improved recognition accuracy and robustness [1]–[3]. However, system-level

bottlenecks remain. Many wearable gesture recognition platforms rely on multi-channel electrical sensing modalities, including surface electromyography, inertial measurement units, and flexible strain, piezoresistive, or capacitive sensor arrays. Such systems typically require analog front-end conditioning and analog-to-digital conversion, followed by inference on an MCU, FPGA, or edge processor. As the number of sensing channels increases with the number of joints and degrees of freedom, readout, digitization, and data movement can dominate latency and power consumption [4], [5].

Inertial measurement unit (IMU) and electromyography (EMG) based approaches can enable effective classification using sensors placed on the wrist or forearm, but they generally require continuous multi-channel sampling and windowed feature extraction [1], [5]. Their system cost therefore scales with sampling bandwidth and channel count, complicating always-on operation under tight power budgets. Flexible resistive or capacitive gloves and electronic skins offer dense spatial sensing for fine-grained finger articulation, yet they similarly require multi-point readout and careful calibration to maintain stability and repeatability [6]–[8]. Consequently, the dominant overhead is often not neural network inference per se, but rather sensing readout, digitization, and the associated data movement.

Optical wearable sensors provide an alternative interface with electromagnetic immunity, mechanical compliance, and low-loss transmission over distance. Fiber Bragg grating (FBG) gloves have demonstrated real-time hand posture monitoring by multiplexing sensing sites along fibers [9]. Related studies report gesture recognition using wearable FBG structures embedded in soft encapsulants [10], [11]. Surveys further show that intensity-based and interferometric fiber sensors can be engineered for joint and motion monitoring with high sensitivity [12], [13]. In most optical sensing implementations, however, the measured signals are still converted and digitized through multi-channel photodetection or specialized demodulation hardware, limiting how much ADC and readout overhead can be reduced at the system level.

In parallel, programmable photonic integrated circuits (PICs) have matured as a platform for analog linear computing. Universal linear transformations can be implemented using interferometer meshes based on established decompositions and scalable designs [14], [16]. Reconfigurable meshes have been used as trainable linear operators for channel mixing, filtering, and matrix multiplication [17], [18], motivating extensive work on photonic neural and neuromorphic computing in which optical linear layers provide high throughput and low latency for feature extraction and inference acceleration [19]–[21].

This work was supported by National Natural Science Foundation of China (NSFC) under Grant (62175103, 61775094, 62341408, 62175100), Fundamental Research Funds for the Central Universities (2024300421, 2024300447, 0213-14380265), Jiangsu Innovation Teams and AI & AI for Science Project of Nanjing University. (Corresponding authors: Yixin Zhang and Ningmu Zou.)

Kangjian Di, Jiashu Li, Shengdi Chen, and Silin Chen are with the School of Integrated Circuits, Nanjing University, Suzhou 215163, China, and also with the Ministry of Education Key Laboratory of Intelligent Optical Sensing and Manipulation, Nanjing University, Nanjing 210023, China.

Fuhao Yu, Nasheeta Mazhar, Wei Jiang, Xuping Zhang, and Yixin Zhang are with the College of Engineering and Applied Sciences, Nanjing University, Nanjing 210023, China, and also with the Ministry of Education Key Laboratory of Intelligent Optical Sensing and Manipulation, Nanjing University, Nanjing 210023, China. (email: yixin@nju.edu.cn)

Ningmu Zou is with the School of Integrated Circuits, Nanjing University, Suzhou 215163, China, and also with the Ministry of Education Key Laboratory of Intelligent Optical Sensing and Manipulation, Nanjing University, Nanjing 210023, China, and also with the Interdisciplinary Research Center for Future Intelligent Chips (Chip-X), Nanjing University, Suzhou 215163, China. (email: nzou@nju.edu.cn)

Kangjian Di and Jiashu Li contributed equally to this work.

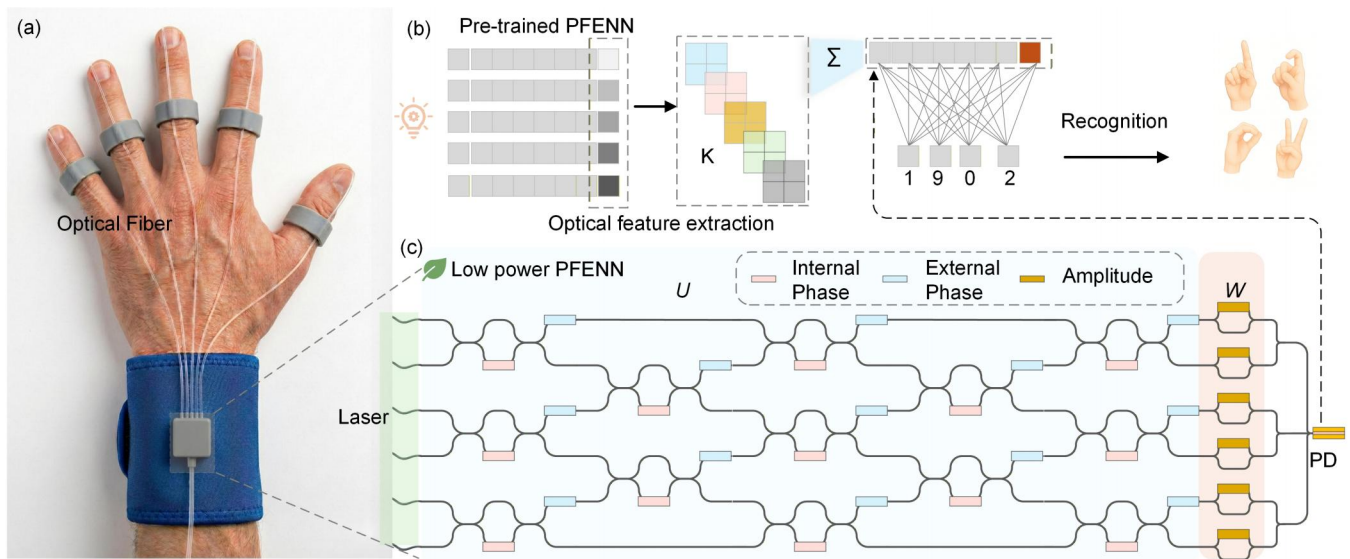


Fig. 1. Overview of the proposed all-optical sensing and learnable photonic processing pipeline for wearable hand gesture recognition. (a) All-optical gesture sensing concept. Standard optical fibers are strapped to finger joints, where joint bending induces curvature variations and calibrated macro-bending loss, thereby encoding gestures directly as optical intensity time series. (b) Pre-trained PFENN electronic back-end for classification. The fused optical output is detected by a photodiode (PD) and fed into a compact fully connected network to perform gesture recognition. (c) PFENN chip architecture. Multiple optical input channels are processed by a $N \times N$ MZI mesh with on-chip phase shifters and modulators that implement a trainable linear operator for feature extraction, followed by passive optical combining realized by multiple couplers to compress all channels into a single readout path.

Recent near-sensor photonic edge computing further illustrates how photonics can move computation closer to the data source to reduce data movement [22].

Here, we report a photonic feature extraction neural network (PFENN) for wearable gesture recognition that integrates passive, fiber-based optical sensing with a learnable photonic preprocessing front end. Gestures are encoded directly in the optical domain via calibrated macro-bending loss in standard single-mode fibers attached to finger joints, producing four optical intensity time series that are delivered to a programmable silicon photonic chip without electrical transduction at the sensing interface. On chip, a learnable photonic linear filter bank is realized by a Mach-Zehnder interferometer (MZI) mesh with quasi-static diagonal weighting during inference, and the processed channels are passively combined into a single optical output for readout. A single photodiode and a compact electronic classifier then complete recognition. By aligning intensity-based fiber sensing with a learnable photonic linear filter bank, our approach enables optical-domain preprocessing before photodetection and structurally reduces multi-channel electrical readout and ADC requirements.

II. METHODS

A. System overview and signal flow

Figure 1 summarizes the proposed optical sensing and photonic preprocessing pipeline for wearable gesture recognition. As shown in Fig. 1a, finger-joint bending modulates the curvature of optical fibers attached to the joints, and the resulting calibrated macro-bending loss encodes each gesture directly as a time-varying optical intensity stream. The multiple optical channels are delivered to a programmable PIC that implements the proposed PFENN. The PFENN architecture, depicted in

Fig. 1c, consists of a reconfigurable $N \times N$ MZI mesh that performs a trainable linear transformation, followed by a diagonal weighting stage realized with multiple Mach-Zehnder modulators (MZMs). The weighted outputs are then fused by multiple couplers into a single optical readout. This fused signal is detected by a single photodiode and subsequently classified by a compact fully-connected electronic network (FCNet), illustrated in Fig. 1b. In practice, we first model the PFENN and pre-train it; the trained unitary matrix (from the MZI mesh) and diagonal weighting matrix (from the MZMs) are then mapped onto the physical chip via calibrated phase-shifter settings. Thus, the pipeline performs low-latency, optical-domain feature extraction before final electronic classification.

B. All-optical gesture sensing via macro-bending loss

A bent single-mode fiber can be modeled as a weakly guiding waveguide whose axis follows a circular arc of radius R . Bending perturbs the guided mode and induces radiation leakage because a fraction of the modal field extends into the cladding and becomes weakly confined. This leakage produces an attenuation coefficient that depends strongly on the bending radius.

Figure 2 illustrates the macro-bending-loss sensing principle and its calibration using a standard single-mode fiber. When a finger joint bends, the fiber curvature increases and the effective bending radius R decreases, introducing macro-bending-induced attenuation and reducing the transmitted optical intensity. To ensure that the fiber bending curvature faithfully follows the joint articulation, the fiber is tightly strapped across the finger joint using adhesive tape at both ends. This constrains the fiber's deformation path, forcing it to conform to the

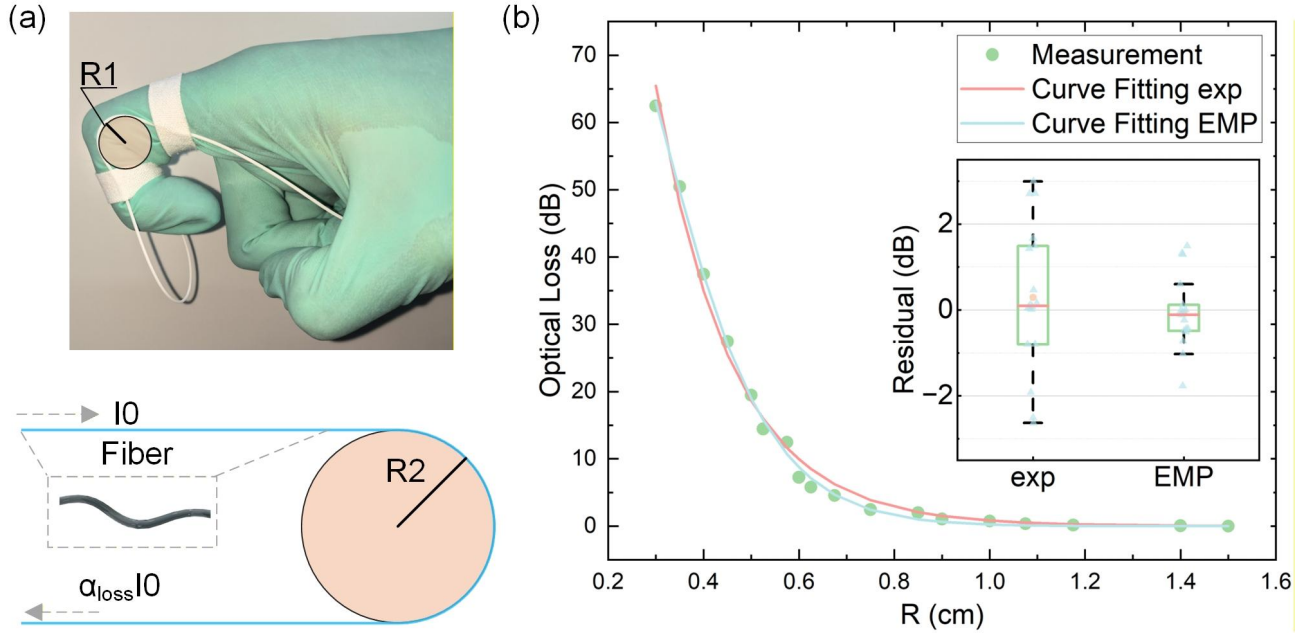


Fig. 2. Principle of all-optical gesture sensing via macro-bending loss. (a) Schematic illustration of fiber macro-bending with two representative curvature radii R_1 and R_2 . (b) Measured optical loss (dB) versus bending radius R (cm), together with curve fitting results using an exponential model (“exp”) and an empirical model (“EMP”). Inset: residual distributions (dB) for the two fitting models, demonstrating the fitting accuracy.

finger surface and thus establishing a repeatable relationship between joint angle and macro-bending loss. Figure 2b plots the measured optical loss (dB) as a function of bending radius R (cm). For each radius, we recorded the transmitted power $P(R)$ in dBm and defined the relative macro-bending loss as $L(R) = P_{\text{ref}}(\text{dBm}) - P(R)(\text{dBm})$, where P_{ref} is the maximum measured power across all radii. The loss increases rapidly as R decreases, providing a sensitive and monotonic transduction mechanism for joint motion. Two fitting models are shown: the RMSE of the exponential model is 1.653 dB, and the RMSE of the empirical model is 0.771 dB.

Macro-bending loss in single-mode fibers often exhibits an approximately exponential dependence on bending radius over practical curvature ranges [23]. One model is

$$L_{\text{exp}}(R) = A \exp(-BR). \quad (1)$$

We also use an empirical model,

$$L_{\text{emp}}(R) = C_1 R^{C_2} \exp(-C_3 R), \quad (2)$$

where the additional power-law term increases flexibility to capture deviations due to coating effects, finite bend length, transition loss, and non-ideal mounting [24].

In our experiments, the sensing element is a single-mode fiber, so the guided field at the sensing wavelength is dominated by the fundamental mode. This suppresses multimode interference and yields a stable and repeatable relationship between curvature and transmitted power. During gesture acquisition, joint motion produces a time-varying effective radius $R(t)$ and thus a time-varying loss $L(t)$, encoding each gesture into an optical intensity time series. The fitted calibration model provides a compact mapping from mechanical curvature to optical attenuation for downstream photonic processing.

C. Principle of PFENN

Our PFENN consists of an MZI mesh followed by MZMs and an N -to-1 passive combining network. A key reason MZI meshes are widely used as programmable linear processors is that a single 2×2 MZI can implement a general unitary transformation on two optical modes when equipped with tunable phase shifters [15]. In a common model, a 2×2 MZI comprises two ideal 50:50 couplers and two controllable phase shifts. Let ρ denote the power coupling ratio of each directional coupler (for a 3 dB coupler, $\rho = 0.5$). The complex transfer matrix of a reconfigurable MZI can be written as

$$\mathbf{D}_{\text{MZI}} = \begin{bmatrix} e^{j\phi} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{\rho} & j\sqrt{1-\rho} \\ j\sqrt{1-\rho} & \sqrt{\rho} \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & e^{j\theta} \end{bmatrix} \begin{bmatrix} \sqrt{\rho} & j\sqrt{1-\rho} \\ j\sqrt{1-\rho} & \sqrt{\rho} \end{bmatrix}. \quad (3)$$

For $\rho = 0.5$, an equivalent closed form is

$$\mathbf{D}_{\text{MZI}} = j e^{j\theta/2} \begin{bmatrix} e^{j\phi} (-\sin \frac{\theta}{2}) & e^{j\phi} \cos \frac{\theta}{2} \\ \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{bmatrix}. \quad (4)$$

An $N \times N$ MZI mesh is constructed by cascading and interconnecting multiple such 2×2 blocks in a structured topology (e.g., triangular or rectangular) [14], [16]. By appropriately setting the phase shifters, the mesh implements a desired $N \times N$ unitary (or approximately unitary) linear operator. We denote the effective transfer matrix by $\mathbf{U} \in \mathbb{C}^{N \times N}$,

$$\mathbf{U} = [u_{ij}]_{N \times N}. \quad (5)$$

Let the complex field amplitudes at the N input ports be

$$\mathbf{x}(t) = [x_1(t), x_2(t), \dots, x_N(t)]^T \in \mathbb{C}^N. \quad (6)$$

The mesh performs complex matrix multiplication,

$$\mathbf{y}(t) = \mathbf{U}\mathbf{x}(t), \quad (7)$$

$$y_k(t) = \sum_{j=1}^N u_{kj}x_j(t), \quad k = 1, \dots, N. \quad (8)$$

If all channels were mutually coherent at the same wavelength, the output power at port k would be $P_k(t) = |y_k(t)|^2$ and would include cross terms. In our implementation, input port j carries wavelength λ_j (or an effectively incoherent source), so different channels do not interfere. For a fixed output port k and wavelength λ_j injected at input port j , the complex field contribution at port k is $u_{kj}x_j(t)$, and the corresponding optical power is

$$P_k^{(j)}(t) = |u_{kj}x_j(t)|^2 = |u_{kj}|^2|x_j(t)|^2. \quad (9)$$

Since the wavelengths are mutually incoherent, the total power at output port k is the sum over wavelengths,

$$P_k(t) = \sum_{j=1}^N P_k^{(j)}(t) = \sum_{j=1}^N |u_{kj}|^2|x_j(t)|^2. \quad (10)$$

After the $N \times N$ mesh, each output channel passes through an MZM. We model this stage as a diagonal weighting operator,

$$\mathbf{W} = \text{diag}(w_1, w_2, \dots, w_N), \quad \mathbf{z}(t) = \mathbf{W}\mathbf{y}(t), \quad (11)$$

where w_i represents the effective complex transmission of the i -th MZM at its operating bias (equivalently, $|w_i|^2$ acts as a power weight).

The N weighted channels are compressed into a single optical output using a passive combining network composed of cascaded 2×1 couplers (e.g., an MMI tree). For each wavelength λ_j , the combining network is a linear map. We represent the entire N -to-1 combiner by a fixed combining vector $\mathbf{c} \in \mathbb{C}^N$,

$$y_{\text{out}}(t) = \mathbf{c}^T \mathbf{z}(t) = \mathbf{c}^T \mathbf{W} \mathbf{U} \mathbf{x}(t), \quad (12)$$

where $\mathbf{c}$ absorbs the tree topology, splitting ratios, phase offsets, and port-dependent imbalances in practice (for an ideal balanced combiner, one may target $|c_i| \approx 1/\sqrt{N}$).

Under WDM, only $x_j(t)$ is present at wavelength λ_j . Propagating wavelength-by-wavelength through the PIC and combiner, the fused output field contribution at λ_j can be written as

$$y_{\text{out}}^{(j)}(t) = h_j x_j(t), \quad h_j = \sum_{i=1}^N c_i w_i u_{ij}. \quad (13)$$

The photodiode measures optical power. Since the wavelengths are mutually incoherent, the detected power equals the sum of powers over wavelengths,

$$P_{\text{out}}(t) = \sum_{j=1}^N |y_{\text{out}}^{(j)}(t)|^2 = \sum_{j=1}^N |h_j|^2 |x_j(t)|^2. \quad (14)$$

Accordingly, propagation and interference are computed per wavelength through the PIC and combiner, and the resulting powers are summed at the detector. Equivalently, by defining

the input power vector as $\mathbf{p}(t) = [|x_1(t)|^2, \dots, |x_N(t)|^2]^T$ and the effective power-domain weight vector as $\mathbf{g} = [|h_1|^2, \dots, |h_N|^2]^T$, the experimentally realized PFENN under mutually incoherent WDM operation implements a trainable power-domain operator,

$$P_{\text{out}}(t) = \mathbf{g}^T \mathbf{p}(t). \quad (15)$$

Therefore, the MZI mesh, MZM weighting stage, and passive combiner coherently process each individual wavelength, while the single photodiode sums the powers of mutually incoherent wavelength components.

D. Dataset collection of gesture signals

Gesture data were collected following the attachment scheme illustrated in Fig. 3a. Due to the limited number of available photonic input channels, fibers were attached to the thumb, index, middle, and ring fingers. Thirteen distinct gestures were performed; therefore, the gesture set does not rely on motion of the little finger. The “Open” gesture in the top-left corner represents the reset state of the hand postures. Although gestures such as “A”, “E”, and “S” appear visually similar, their optical signals (Figs. 3b and 3c) exhibit discernible differences. In particular, the distinction between “S” and “A” primarily reflects different degrees of thumb flexion, whereas the difference between “E” and “S” arises from variations in the flexion of both the thumb and the other three fingers.

In Fig. 3b, the four laser sources were configured to emit equal optical power. In practice, each channel was calibrated to -3 dBm for consistency. The pink-shaded regions denote transitions from the open (rest) state to the target gesture, while the green-shaded regions correspond to steady-state open-hand periods.

To decouple software pre-training from hardware evaluation and to enable a controlled comparison between trained and non-trained photonic configurations, three datasets under the same 13-gesture protocol were constructed. Each dataset contains 200 samples per gesture (2600 samples per dataset). All experiments are conducted with a single subject. Unless otherwise stated, a 60%/40% train/test split is used within each dataset.

Dataset A. Dataset A contains four-channel optical intensity time series acquired from the all-optical sensing interface *before* the signals are injected into the photonic chip. The four wavelength channels were measured using four photodiodes. Dataset A is used to pre-train the PFENN model in software.

Dataset B. After software pre-training, the learned photonic parameters are mapped to chip control currents using measured device responses and calibration. Dataset B is collected by routing the four sensing channels into the PFENN chip while loading the trained photonic weights. The chip produces a fused output that is detected by a single photodiode.

Dataset C. Dataset C is collected using the same optical routing and measurement chain as Dataset B, except that the PFENN chip is configured with random phase settings (random $\mathbf{U}$ and/or random $\mathbf{W}$, consistent with the baseline definition).

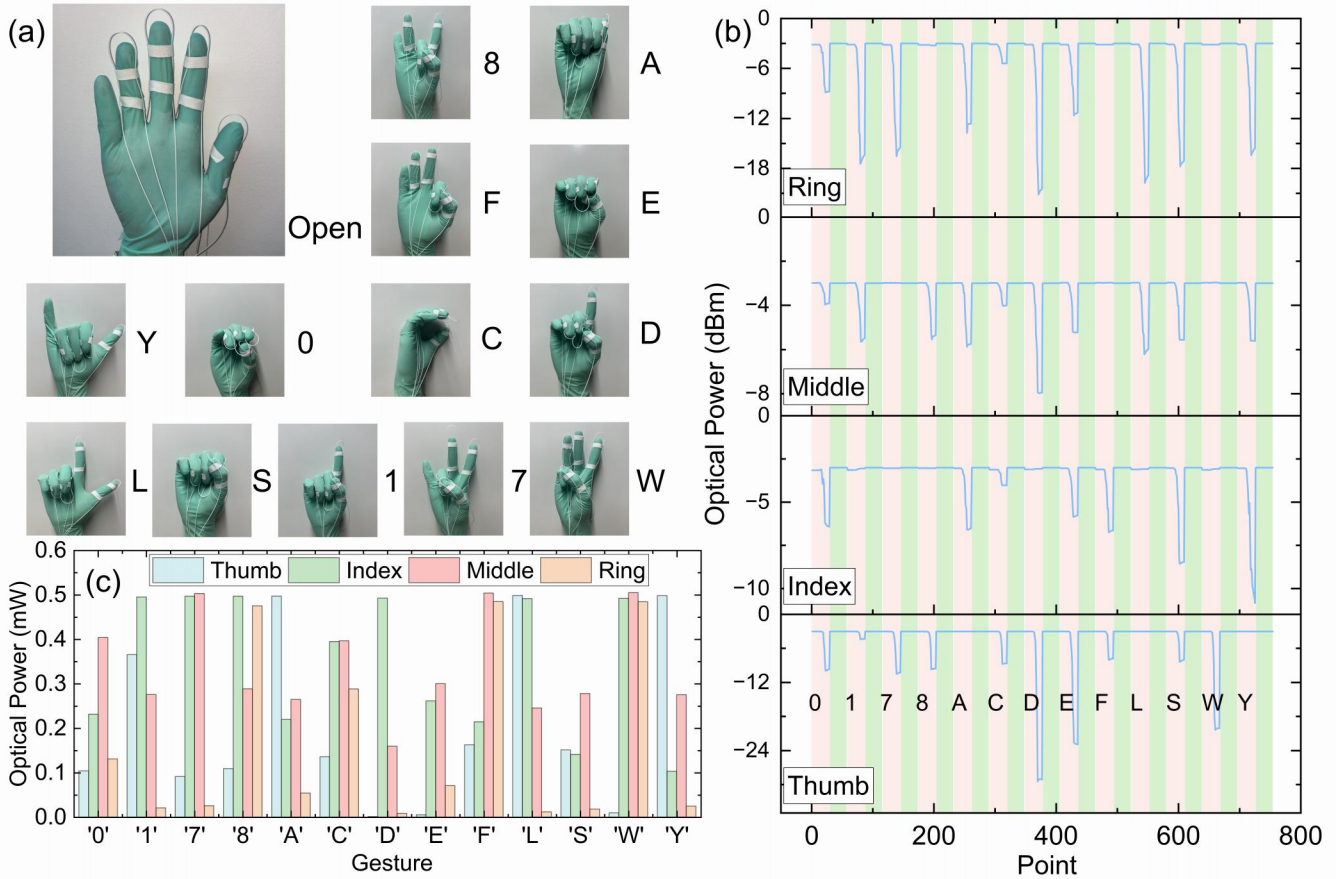


Fig. 3. Collection of gesture signals via the all-optical sensing interface. (a) Gesture categories performed in the experiment. (b) Corresponding curvature waveforms for different hand gestures during motion. (c) Baseline signals in the static (no-motion) state for each gesture.

TABLE I
ELECTRONIC FCNET ARCHITECTURE AND TRAINING
HYPERPARAMETERS.

Item	Setting
Input feature dimension	30
Electronic classifier	FCNet
Network architecture	FC(30, 30)
	ReLU
	FC(30, 13)
Number of gesture classes	13
Loss function	Cross-entropy loss
Batch size	32
Epochs	500
Learning rate for electronic parameters	0.002
Learning rate for optical parameters	0.02
Optical-parameter update	Forward difference

For a fair comparison between Datasets B and C, the same FCNet architecture was trained separately on the fused outputs of Dataset B and Dataset C using the same 60%/40% split. The full FCNet architecture and key training hyperparameters are summarized in Table I. The FCNet uses ReLU as the nonlinear activation function between the fully connected layers. All results are reported for a single subject under a fixed train/test split, serving as an initial experimental validation of the proposed photonic front end; evaluating inter-subject generalization is beyond the scope of this study.

III. EXPERIMENTAL SETUP AND RESULTS DISCUSSION

We implemented the experimental setup shown in Fig. 4 to implement the proposed all-optical wearable gesture recognition system. The PFENN chip serves as the photonic front end for feature extraction from the gesture signals. A micrograph of the fabricated chip is presented in Fig. 4a. The chip was fabricated on a commercial silicon-on-insulator (SOI) wafer featuring a 220-nm-thick top silicon layer and a 3- μm -thick buried oxide layer. The optical waveguides are 220-nm-thick silicon strip waveguides. Grating couplers for vertical coupling are implemented as 70-nm shallow-etched, fan-shaped structures. A cladding oxide layer was deposited over the waveguides, followed by TiN heater deposition. Each heater is 140 μm long and 4 μm wide.

Light is coupled into and out of the chip through grating couplers on the left side. In PFENN operation, four input grating couplers inject light into a 4×4 MZI mesh implementing the programmable unitary transformation $\mathbf{U}$. The shaded regions in the schematic denote phase-shifter sections, categorized as internal phase shifters (within the mesh), external phase shifters, compensation phase shifters, and phase shifters dedicated to the MZMs. After processing by $\mathbf{U}$, the light is routed to the diagonal weighting stage $\mathbf{W}$ realized by the MZMs. The weighted channels are then passively combined by an MMI coupler tree and output through a single grating

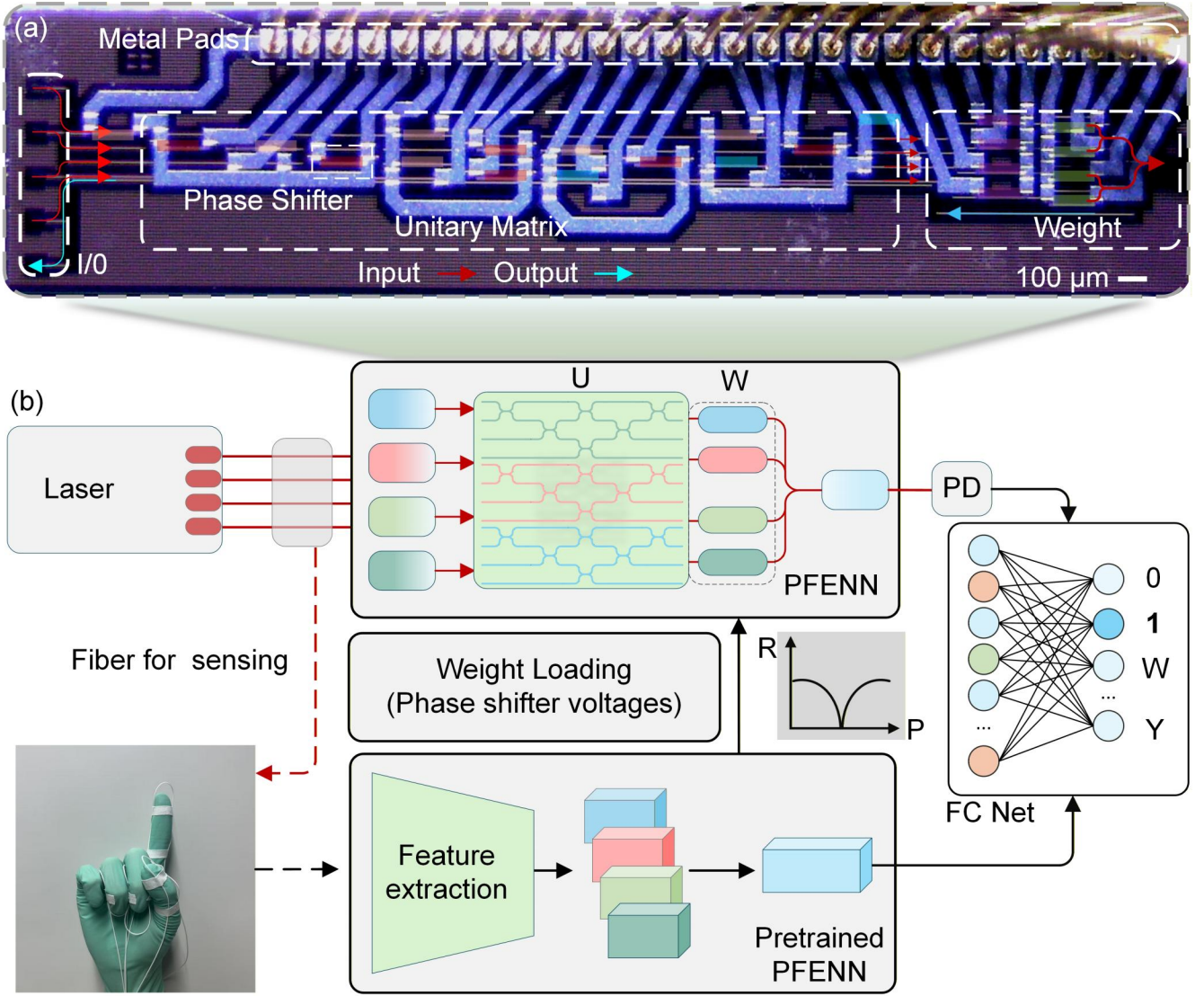


Fig. 4. Experimental setup for all-optical wearable gesture recognition based on the PFENN. (a) Microscope image of the fabricated PFENN PIC. (b) Schematic diagram of the proposed all-optical sensing system for wearable hand gesture recognition.

coupler.

The experimental framework is shown in Fig. 4b. While mutually coherent inputs would preserve full phase-sensitive interference across channels, gesture recognition based on curvature-induced intensity modulation benefits from suppressing inter-channel phase sensitivity prior to photonic processing. Therefore, four tunable lasers (Santec TSL510; 1 mW output power; ~ 200 kHz linewidth) at $\lambda_1 = 1548$ nm, $\lambda_2 = 1550$ nm, $\lambda_3 = 1552$ nm, and $\lambda_4 = 1554$ nm were employed. A polarization controller was inserted in each channel before delivery to the sensing glove. The modulated optical signals were then routed into the PFENN chip. A photodetector (HP 81531A; DC to 400 Hz bandwidth) captured the fused output and performed square-law detection to produce a one-dimensional gesture time series.

The optical fibers used in the experiment are standard single-mode fibers. The laser output is connected via a 0.5 m fiber jumper to a 1.5 m sensing-fiber section attached to

the hand and then coupled into the photonic chip through a 0.5 m input fiber. After processing, the chip output is delivered through another 0.5 m fiber jumper to the photodetector, for a total fiber length of approximately 3 m.

Prior to the experiment, a theoretical model of the PIC was established based on Section II-A. Dataset A was mapped into this model to pre-train the PFENN, which involves optimizing the trainable 4×4 unitary operator U and diagonal weighting matrix W . Parameters were updated using the forward-difference algorithm. The model output was then fed to a fully connected network for gesture classification.

The trained parameters for U and W were extracted and mapped to the corresponding control currents of the physical PIC. Figure 5a shows representative time-domain outputs after photonic preprocessing with the trained weights loaded. These fused features exhibit distinct waveform and amplitude patterns across gestures, supporting accurate classification with a compact electronic back-end. Due to fabrication variability,

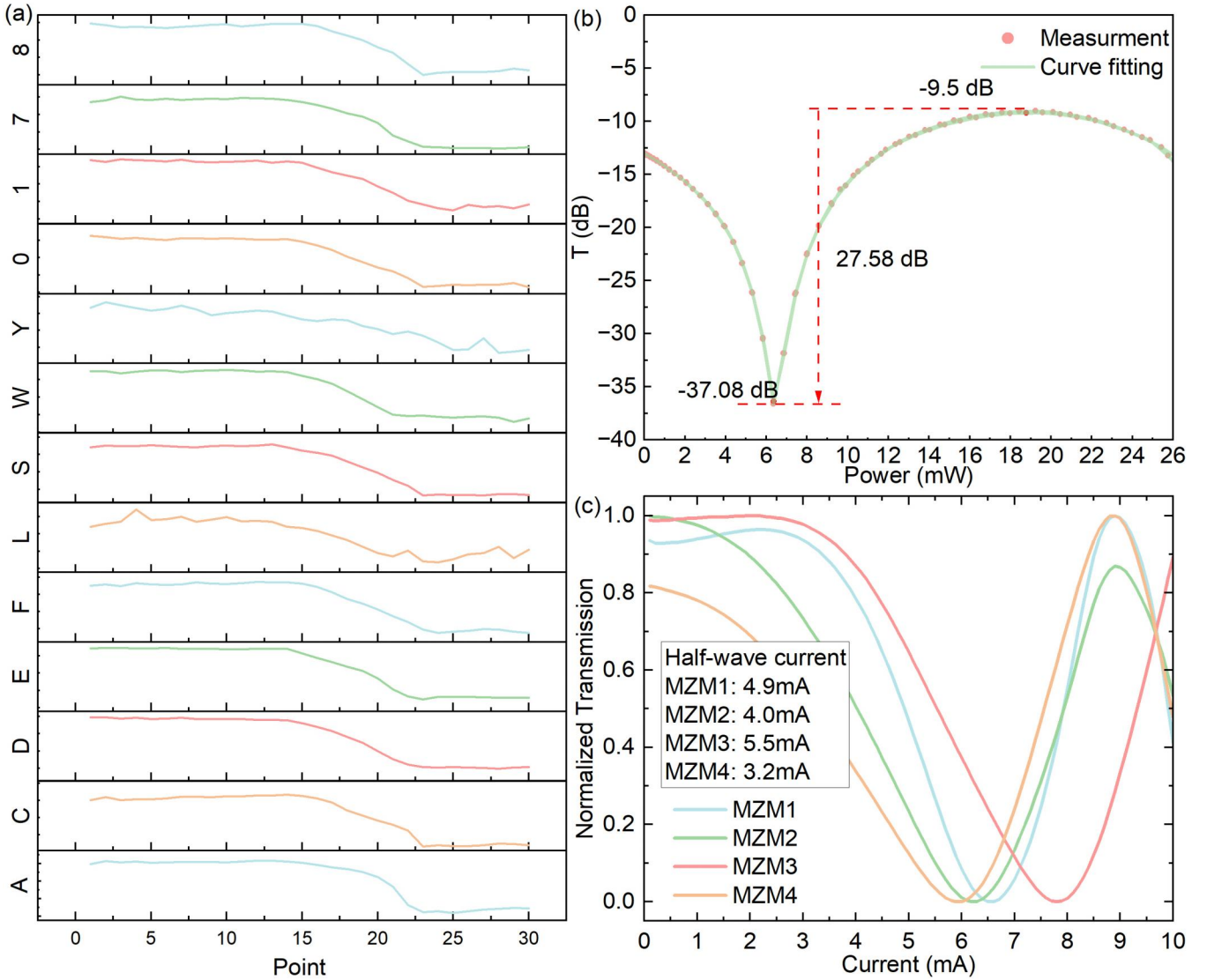


Fig. 5. Characterization of gesture signals and the PFENN chip. (a) Time-domain signals of gesture features extracted by the PFENN with the pre-trained weights loaded. (b) Modulation curves of the MZMs in the PFENN, measured within the same batch. (c) Phase-shift curves represented by the diagonal weighting matrix $\mathbf{W}$ of the PFENN.

we performed phase scanning and calibration to align the PIC response with the pre-trained matrices.

Figures 5b and 5c summarize the measured MZM characteristics. Figure 5b shows a typical MZM modulation curve with an extinction ratio of 27.58 dB. The phase shifters are driven by a current source (NI 9266; 24 kS/s update rate per channel; 16-bit resolution), enabling precise weight loading. Figure 5c shows the phase-shift curves corresponding to the four diagonal weights in $\mathbf{W}$. The measured half-wave currents are 3.2 mA, 4.0 mA, 4.9 mA, and 5.5 mA, respectively. Although the resulting attenuation varies across channels, this non-uniformity can be compensated through electronic calibration. After weight loading, strict matching of absolute optical input power to the pre-training condition was not required because the system response is linear in optical power; however, equal optical power across the four input channels was maintained.

Figure 6 summarizes recognition performance. Figure 6a

compares four configurations: (i) a digital PFENN with an FCNet trained on Dataset A, (ii) the full photonic PFENN with trained photonic weights and an FCNet trained on Dataset B, (iii) a photonic front end initialized with random phases and weights with an FCNet trained on Dataset C, and (iv) an FCNet trained directly on the raw four-channel optical intensity signals acquired before the PFENN chip. The proposed system achieves 96.63% accuracy on the 13-gesture dataset. Compared with the digital PFENN reference accuracy of 98.90%, this corresponds to a 2.27% drop after chip deployment. This gap provides an estimate of the hardware deployment cost, including residual phase-setting errors, device imbalance, and calibration residuals. The confusion matrices in Fig. 6d–f show that most gesture classes are classified with high accuracy. For the pre-trained PFENN, classes “F” and “C” exhibit higher similarity after feature extraction and thus account for a larger fraction of the residual errors. By contrast, the non-trained PFENN provides less discriminative

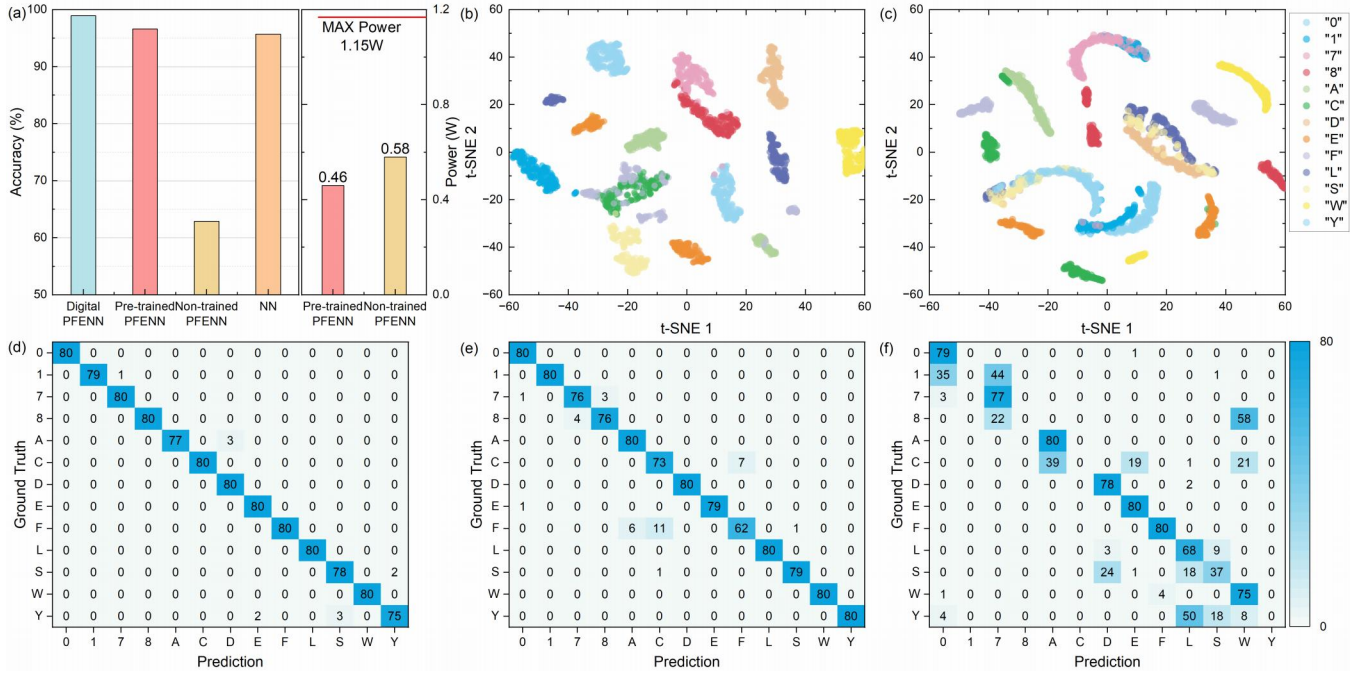


Fig. 6. All-optical gesture recognition results using PFENN. (a) Classification accuracy and on-chip tuning power for four configurations: digital PFENN, pre-trained PFENN, non-trained PFENN, and NN. (b) t-SNE visualization of features extracted by the *pre-trained* PFENN. (c) t-SNE visualization of features extracted by the *non-trained* PFENN. (d) Confusion matrix of the digital PFENN. (e) Confusion matrix of the pre-trained PFENN. (f) Confusion matrix of the non-trained PFENN.

fused features, leading to lower overall accuracy. The raw four-channel NN baseline achieves 95.7% accuracy but requires four photodetectors/readout channels.

To visualize feature separability, Fig. 6b–c shows t-SNE projections of features extracted by the pre-trained and non-trained PFENN, respectively. The trained photonic front end yields more compact and separable clusters, consistent with improved class separability prior to electronic classification. Representative fused waveforms after photonic preprocessing are shown in Fig. 5a.

Digital PFENN serves as a software reference that isolates the algorithmic benefit of the learned photonic operator from hardware nonidealities (e.g., phase-setting errors and device imbalance), whereas pre-trained PFENN quantifies performance after weight mapping and calibration.

A. Latency analysis

The physical propagation latency from sensing to photonic processing is dominated by light propagation in the fiber. For total fiber length $L_{\text{fiber}} \approx 3$ m and silica group index $n_g \approx 1.46$, the propagation delay is

$$\tau_{\text{fiber}} \approx \frac{n_g L_{\text{fiber}}}{c} \approx 15 \text{ ns}. \quad (16)$$

The on-chip waveguide length is approximately $4000 \mu\text{m}$, corresponding to a sub-10-ps to tens-of-ps delay (depending on on-chip group index), which is negligible relative to the fiber delay.

In our laboratory prototype, the photodetector/transimpedance front end has a bandwidth of ~ 400 Hz. Approximating the receiver as a first-order low-pass filter, the

corresponding time constant is $\tau_{\text{PD}} \approx 1/(2\pi B) \approx 0.40$ ms, and the 10–90% rise time is $t_r \approx 2.2\tau_{\text{PD}} \approx 0.88$ ms. Therefore, even when including the PD/TIA response, the end-to-end physical latency from optical sensing to photonic preprocessing remains on the sub-millisecond scale, which is far below typical gesture dynamics and windowing times (tens to hundreds of milliseconds) required for robust recognition.

The reported sensing-to-photonic-processing latency is therefore on the order of ~ 15 ns in terms of pure optical propagation, excluding PD/TIA response, buffering, and gesture windowing/segmentation latency. Notably, the passive fiber link can be extended to support operation in electromagnetically harsh environments by relocating the electronics away from the sensing site; the added latency remains small because it scales only as $\Delta\tau \approx (n_g/c)\Delta L \approx 4.9$ ns/m. For example, extending the fiber by 100 m introduces only $\sim 0.5 \mu\text{s}$ additional delay. Meanwhile, the optical attenuation of standard silica fiber is sufficiently low over such distances, making longer-reach sensing practical without materially impacting the real-time requirement of gesture recognition.

B. Electrical power consumption

Electrical power in the photonic front end is dominated by static tuning of thermo-optic phase shifters and MZM biases implemented as resistive heaters. The chip contains $N_h = 23$ heaters with nominal resistance $R \approx 500 \Omega$. Under the safety constraint $I \leq 10$ mA per heater, a conservative upper bound on heater power is

$$P_{\text{max}} = N_h R I_{\text{max}}^2 = 23 \times 500 \times 0.01^2 \approx 1.15 \text{ W}. \quad (17)$$

In experiments, the measured total heater power after calibration and weight loading is 0.46 W for the trained configuration (Dataset B) and 0.58 W for the random-phase configuration (Dataset C). These values are comparable because both configurations tune the same set of heaters over similar current ranges, and both remain below the conservative bound of 1.15 W. Unless otherwise noted, reported power corresponds to on-chip electrical tuning power and excludes external laser, PD, and instrument power.

IV. CONCLUSION

We demonstrated an optical-domain gesture sensing and preprocessing pipeline that encodes finger motion as wavelength-multiplexed optical intensity streams via calibrated macro-bending loss in standard single-mode fibers, avoiding electrical transduction and digitization at the sensing interface. Four wavelength channels are processed by a programmable silicon photonic chip implementing a learnable linear filter bank based on a 4×4 MZI mesh with static diagonal weighting and passive combining, followed by a single photodiode and a compact electronic classifier. Using a training-to-deployment workflow that maps software-trained weights onto calibrated on-chip responses, the system achieves 96.63% accuracy on a 13-gesture, single-subject dataset (200 samples per class). The physical propagation latency from sensing to photonic processing is ~ 15 ns for an approximately 3 m fiber link. The on-chip tuning power is 0.46 W (trained) and 0.58 W (random), both below a conservative 1.15 W upper bound. Together, these results illustrate a practical approach to low-overhead near-sensor wearable intelligence by combining passive fiber sensing with a learnable photonic linear filter bank.

REFERENCES

- [1] A. Moin, A. Zhou, A. Rahimi, A. Menon, S. Benatti, G. Alexandrov, S. Tamakloe, J. Ting, N. Yamamoto, Y. Khan, F. Burghardt, L. Benini, A. C. Arias, and J. M. Rabaey, "A wearable biosensing system with in-sensor adaptive machine learning for hand gesture recognition," *Nat. Electron.*, vol. 4, no. 1, pp. 54–63, 2020.
- [2] K. R. Pyun, K. Kwon, M. J. Yoo, K. K. Kim, D. Gong, W.-H. Yeo, S. Han, and S. H. Ko, "Machine-learned wearable sensors for real-time hand-motion recognition: toward practical applications," *Natl. Sci. Rev.*, vol. 11, no. 2, p. nwad298, 2024.
- [3] R. Tchanchane, H. Zhou, S. Zhang, and G. Alici, "A Review of Hand Gesture Recognition Systems Based on Noninvasive Wearable Sensors," *Adv. Intell. Syst.*, vol. 5, no. 10, p. 2300207, 2023.
- [4] H. Jeon, H. Choi, D. Noh, T. Kim, and D. Lee, "Wearable Inertial Sensor-Based Hand-Guiding Gestures Recognition Method Robust to Significant Changes in the Body-Alignment of Subject," *Mathematics*, vol. 10, no. 24, p. 4753, 2022.
- [5] U. Cote-Allard, C. L. Fall, A. Drouin, A. Campeau-Lecours, C. Gosselin, K. Glette, F. Laviolette, and B. Gosselin, "Deep Learning for Electromyographic Hand Gesture Signal Classification Using Transfer Learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 4, pp. 760–771, 2019.
- [6] Y. Dong, J. Liu, and W. Yan, "Dynamic Hand Gesture Recognition Based on Signals From Specialized Data Glove and Deep Learning Algorithms," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–14, 2021.
- [7] T. Wang, Y. Zhao, and Q. Wang, "Hand Gesture Recognition With Flexible Capacitive Wristband Using Triplet Network in Inter-Day Applications," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 2876–2885, 2022.
- [8] Q. Wang, M. Li, P. Guo, L. Gao, L. Weng, and W. Huang, "Shape-position perceptive fusion electronic skin with autonomous learning for gesture interaction," *Microsyst. Nanoeng.*, vol. 10, no. 1, p. 103, 2024.
- [9] A. F. Da Silva, A. F. Goncalves, P. M. Mendes, and J. H. Correia, "FBG Sensing Glove for Monitoring Hand Posture," *IEEE Sens. J.*, vol. 11, no. 10, pp. 2442–2448, 2011.
- [10] K. Xiao, Z. Wang, Y. Ye, C. Teng, and R. Min, "PDMS-embedded wearable FBG sensors for gesture recognition and communication assistance," *Biomed. Opt. Express*, vol. 15, no. 3, p. 1892, 2024.
- [11] Z. Sun, "Wearable glove gesture recognition based on fiber Bragg grating sensing using genetic algorithm-back propagation neural network," *Opt. Fiber Technol.*, vol. 87, p. 103874, 2024.
- [12] Z. Luo, J. Xu, X. Zhou, Z. Wang, Y. Liu, S. Wang, S. Yuan, H. Zhang, B. He, J. Xin, H. Wang, W. Zhang, Z. Lin, and L. Wei, "Advances in Fiber-Based Wearable Sensors with Machine Learning," *Adv. Devices Instrum.*, vol. 5, p. 0054, 2024.
- [13] A. Leal-Junior, J. Silva, L. Macedo, A. Marchesi, S. Morau, J. Valentino, F. Valentim, and M. Costa, "The role of optical fiber sensors in the new generation of healthcare devices: a review," *Sens. Diagn.*, vol. 3, no. 7, pp. 1135–1158, 2024.
- [14] M. Reck, A. Zeilinger, H. J. Bernstein, and P. Bertani, "Experimental realization of any discrete unitary operator," *Phys. Rev. Lett.*, vol. 73, no. 1, pp. 58–61, 1994.
- [15] F. Shokraneh, M. S. Nezami, and O. Liboiron-Ladouceur, "Theoretical and Experimental Analysis of a 4×4 Reconfigurable MZI-Based Linear Optical Processor," *J. Lightw. Technol.*, vol. 38, no. 6, pp. 1258–1267, 2020.
- [16] W. R. Clements, P. C. Humphreys, B. J. Metcalf, W. S. Kolthammer, and I. A. Walmsley, "Optimal design for universal multiport interferometers," *Optica*, vol. 3, no. 12, p. 1460, 2016.
- [17] D. Pérez, I. Gasulla, P. Das Mahapatra, and J. Capmany, "Principles, fundamentals, and applications of programmable integrated photonics," *Adv. Opt. Photon.*, vol. 12, no. 3, p. 709, 2020.
- [18] N. C. Harris, J. Carolan, D. Bunandar, M. Prabhu, M. Hochberg, T. Baehr-Jones, M. L. Fanto, A. M. Smith, C. C. Tison, P. M. Alsing, and D. Englund, "Linear programmable nanophotonic processors," *Optica*, vol. 5, no. 12, p. 1623, 2018.
- [19] Y. Shen, N. C. Harris, S. Skirlo, M. Prabhu, T. Baehr-Jones, M. Hochberg, X. Sun, S. Zhao, H. Larochelle, D. Englund, and M. Soljačić, "Deep learning with coherent nanophotonic circuits," *Nat. Photonics*, vol. 11, no. 7, pp. 441–446, 2017.
- [20] A. N. Tait, T. F. De Lima, E. Zhou, A. X. Wu, M. A. Nahmias, B. J. Shastri, and P. R. Prucnal, "Neuromorphic photonic networks using silicon photonic weight banks," *Sci. Rep.*, vol. 7, no. 1, p. 7430, 2017.
- [21] Q. Yao, Y. Sun, X. Ke, Y. Xie, S. Liu, H. Xiong, H. Li, X. Bai, Y. Shi, and D. Dai, "Parallel Matrix–Vector Multiplication Using a Scalable Multiport Interferometer," *Laser & Photonics Reviews*, p. e01876, 2025.
- [22] Z. Ren, Z. Zhang, Y. Zhuge, Z. Xiao, S. Xu, J. Zhou, and C. Lee, "Near-Sensor Edge Computing System Enabled by a CMOS Compatible Photonic Integrated Circuit Platform Using Bilayer AlN/Si Waveguides," *Nano-Micro Lett.*, vol. 17, no. 1, 2025.
- [23] Q. Wang, G. Farrell, and T. Freir, "Theoretical and experimental investigations of macro-bend losses for standard single mode fibers," *Opt. Express*, vol. 13, no. 12, p. 4476, 2005.
- [24] R. T. Schermer and J. H. Cole, "Improved Bend Loss Formula Verified for Optical Fiber by Simulation and Experiment," *IEEE J. Quantum Electron.*, vol. 43, no. 10, pp. 899–909, 2007.